

Carbon Crunch — Receipt OCR & Confidence-Aware Extraction

ML/Ops Internship Shortlisting Assignment | Technical Report

1. Approach

The system converts heterogeneous receipt images into structured JSON while exposing reliability at both OCR and field level. Preprocessing, OCR, extraction, validation, confidence scoring and aggregation are separated into modular components.

2. Image Preprocessing & OCR

Images are resized for OCR, converted to grayscale, moderately deskewed, denoised and enhanced with CLAHE. A benchmark compares raw, CLAHE and adaptive-threshold variants on representative receipts. Adaptive thresholding increased detected tokens but reduced mean OCR confidence in the sample, so it is not blindly applied as the default. Tesseract 5 supplies word-level confidence.

3. Key Information Extraction

Store/date candidates use position, OCR confidence, business-like keywords and penalties for address/metadata. Total candidates use explicit total keywords, monetary-format validation and contextual position. Tax/GST, subtotal, cash, tender and change lines are down-weighted or excluded. Item extraction is conservative to reduce false positives from IDs, dates and phone numbers.

4. Confidence & Reliability

Every extracted field carries a value, confidence and low-confidence flag. Confidence combines OCR confidence with pattern/context evidence. Where payment and change are available, arithmetic consistency independently supports the selected total. Values below 0.70 are flagged; missing/ambiguous values are represented as null rather than fabricated.

5. Results on Supplied Dataset

Metric	Result
Receipt images processed	371
Receipts with valid total	331
Receipts without valid total	40
High-confidence totals (≥ 0.85)	72
Medium-confidence totals (0.70–0.849)	125
Low-confidence totals (< 0.70)	134
Accepted-candidate spend	188,165.58

The supplied dataset has no ground-truth annotation file, so no unverified end-to-end accuracy percentage is claimed. The submission instead reports preprocessing comparisons, OCR confidence, validation outcomes and explicit uncertainty.

6. Challenges, Edge Cases & Improvements

Challenges include faded/low-contrast receipts, skew, multiple currencies/formats, OCR substitutions, split decimals and totals mixed with tax/cash/change. The system addresses these with preprocessing and contextual validation. Future improvements include layout-aware OCR/document models, learned field classification, currency-aware normalization, human-in-the-loop review and a manually annotated benchmark.